EK-10 Time Estimation and Capacity Planning Reference

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1. Document metadata

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Data status	Factual estimation guidance with illustrative numbers
Main topics	Time estimation; Capacity planning; Team overhead; Buffers
Relevant skills and technologies	Schedule analysis
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Purpose and scope: explain how to convert effort into duration, how capacity checks work and how team size and buffers change schedules.

2. Executive summary

Duration equals effort divided by weekly hours; capacity must cover at least 80 percent of effort. Tables show every fictional job at four weekly commitments.

3. Effort, capacity and duration

Duration in weeks = effort hours / (committed hours per week). Capacity = committed hours per week x weeks. A proposal is supported when capacity is at least 80 percent of effort. Example: 420 hours at 20 hours per week is 21 weeks; at 30 hours per week it is 14 weeks.

4. Duration of each fictional job at different weekly commitments

Job	Effort h	Reference weeks	Weeks at 15 h	20 h	30 h	40 h	Deadline weeks	Type
JOB-001	420	14	28.0	21.0	14.0	10.5	12	soft
JOB-002	200	8	13.3	10.0	6.7	5.0	6	hard
JOB-003	480	14	32.0	24.0	16.0	12.0	10	soft
JOB-004	240	9	16.0	12.0	8.0	6.0	8	hard
JOB-005	380	12	25.3	19.0	12.7	9.5	10	soft
JOB-006	320	11	21.3	16.0	10.7	8.0	9	soft
JOB-007	800	22	53.3	40.0	26.7	20.0	16	soft
JOB-008	120	5	8.0	6.0	4.0	3.0	5	hard
JOB-009	260	9	17.3	13.0	8.7	6.5	8	soft
JOB-010	140	6	9.3	7.0	4.7	3.5	6	soft

Weeks needed at a given commitment come straight from effort divided by hours per week. Compare with the deadline: when even 40 hours per week cannot meet it, the deadline is aggressive for one person.

5. Teams and coordination overhead (illustrative)

With n people, effective capacity = n x hours x (1 - 0.1 x (n - 1)). Example: two people at 20 hours each give 2 x 20 x 0.9 = 36 hours per week instead of 40; three give 3 x 20 x 0.8 = 48. Adding people helps only when work can be split.

6. Calendar time and buffers

- Add 10 percent for holidays and unavailable days over long projects.
- Add a schedule buffer of 10 to 20 percent before external milestones.
- Add time for client review cycles: each review and fix cycle commonly needs several working days.
- Include time to start: availability dates differ (see FP-08).

7. Testing and release time

Testing and defect fixing is about 15 percent of effort in the illustrative model; deployment and handover about 10 percent. Omitting them shortens a proposal on paper but not the real project.

8. Worked example

A client's soft deadline is 12 weeks for a 420-hour job. A freelancer committing 20 hours per week has capacity 240 hours in 12 weeks, 57 percent of effort, below the 80 percent threshold: capacity risk. At 35 hours per week capacity is 420 hours (100 percent). The assistant reports the required weekly commitment instead of calling the plan impossible.

9. Evidence requirements and uncertainty

Hours and durations are estimates. State the assumptions and give a range when possible; do not present a single duration as certain.

10. Summary of key findings

- Duration = effort / hours per week.
- Team overhead reduces capacity gains.
- Report required weekly hours instead of declaring a plan impossible.

11. Keywords and alternative terminology

time estimation; capacity planning; duration; schedule buffer; team overhead; calendar time